

ET AI Hackathon 2026 — Idea Shortlist for Selection

Problem Statement 5: AI-Powered Urban Air Quality Intelligence Submission deadline: July 22, 2026

Recommended Hero Idea

1. Pollution Source Fingerprinting Engine

Instead of just forecasting AQI numbers, explain *why* it's spiking — attribute each spike to a likely cause (traffic, construction dust, stubble burning, festival burning) using real pollutant-ratio signatures (PM2.5:PM10, NO2 co-occurrence, rate of rise, time of day) cross-checked against wind data.

- **Data:** 100% real — CPCB/OpenAQ pollutant data + Open-Meteo wind data
 - **Why it wins:** No other team is likely to attempt attribution without satellite imagery/land-use ML. This does it through defensible domain logic instead — technically honest and demo-friendly.
 - **Effort:** Medium — rule engine + LLM only for the explanation text, not the classification itself
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Strong Supporting Ideas (pick 1-2 to layer on top)

2. Wind-Direction Spread Forecast

Project which downwind zones will be hit next in the following 6 hours using real wind vectors from Open-Meteo. Renders as an animated "plume" on the map.

- **Effort:** Low (reuses hero idea's data pipeline)
- **Demo value:** High — visually distinctive, nobody else will show this

3. Health-Vulnerability Overlay

Overlay real hospital/school locations (OpenStreetMap Overpass API) on the AQI heatmap to flag which specific vulnerable sites are in high-risk zones right now.

- **Effort:** Low — static geodata, one-time pull
- **Demo value:** High — makes the platform feel human/impact-driven, not just technical

4. Cross-City Comparative Intelligence

Compare AQI trend patterns across Delhi, Mumbai, Bengaluru, Chennai to surface "City X's intervention pattern worked, City Y hasn't tried it."

- **Effort:** Medium — needs clean multi-city historical data
 - **Demo value:** Medium — good for judges scoring "Scalability" and "Business Impact"
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Baseline / Must-Have (not differentiators, but required by the problem statement)

- 24-72hr hyperlocal AQI forecast (Prophet or LSTM, trained on real CPCB/OpenAQ history)
- Geospatial heatmap over real station coordinates (Leaflet/Mapbox, IDW interpolation)
- Multi-language citizen advisory (LLM translation layer — Hindi/Kannada/Tamil)

Deliberately Deprioritized / Mock Only

- Enforcement recommendation list — generate plausibly from attribution output, don't over-invest
 - Live traffic feed — no free real-time API exists; use Vahan vehicle registration data as a static density proxy instead, and state this honestly in the demo
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Recommended Build: Hero Idea (#1) + Supporting Ideas (#2 and #3)

This combination gives three distinct, demo-able screens — technical forecast, visual plume animation, human-impact overlay — from one shared real-data pipeline, without overextending the 2-week timeline.